

Homework 4

Joseph Shin

2025-09-12

```
temps = read.table('temps copy.txt', header = TRUE)
temps
```

##	DAY	X1996	X1997	X1998	X1999	X2000	X2001	X2002	X2003	X2004	X2005	X2006
## 1	1-Jul	98	86	91	84	89	84	90	73	82	91	93
## 2	2-Jul	97	90	88	82	91	87	90	81	81	89	93
## 3	3-Jul	97	93	91	87	93	87	87	87	86	86	93
## 4	4-Jul	90	91	91	88	95	84	89	86	88	86	91
## 5	5-Jul	89	84	91	90	96	86	93	80	90	89	90
## 6	6-Jul	93	84	89	91	96	87	93	84	90	82	81
## 7	7-Jul	93	75	93	82	96	87	89	87	89	76	80
## 8	8-Jul	91	87	95	86	91	89	89	90	87	88	82
## 9	9-Jul	93	84	95	87	96	91	90	89	88	89	84
## 10	10-Jul	93	87	91	87	99	87	91	84	89	78	84
## 11	11-Jul	90	84	91	82	96	90	84	84	90	83	90
## 12	12-Jul	91	88	86	77	93	90	77	86	89	86	91
## 13	13-Jul	93	86	88	73	91	86	82	87	91	84	91
## 14	14-Jul	93	90	87	81	93	82	88	84	91	87	91
## 15	15-Jul	82	91	91	81	93	82	91	86	84	84	91
## 16	16-Jul	91	91	87	86	93	84	93	88	84	85	91
## 17	17-Jul	96	89	90	82	91	87	93	88	84	89	93
## 18	18-Jul	95	89	91	87	97	88	93	88	87	90	93
## 19	19-Jul	96	89	95	88	100	90	93	88	84	89	96
## 20	20-Jul	99	90	91	90	99	87	91	88	88	89	93
## 21	21-Jul	91	89	91	90	93	84	95	89	89	90	93
## 22	22-Jul	95	84	89	91	96	87	91	86	89	91	91
## 23	23-Jul	91	87	91	93	87	90	89	81	93	91	86
## 24	24-Jul	93	88	91	93	82	84	87	82	95	90	87
## 25	25-Jul	84	89	86	91	75	82	84	84	89	92	88
## 26	26-Jul	84	89	88	93	82	88	86	87	87	94	93
## 27	27-Jul	82	91	80	93	88	90	89	87	84	92	95
## 28	28-Jul	79	91	88	93	91	84	91	89	89	90	96
## 29	29-Jul	90	89	89	93	89	89	91	88	87	83	91
## 30	30-Jul	91	88	90	97	87	89	88	84	89	78	91
## 31	31-Jul	87	72	86	99	86	87	90	88	90	84	94
## 32	1-Aug	86	80	86	96	86	84	93	84	91	82	95
## 33	2-Aug	90	84	82	93	81	84	91	84	90	86	95
## 34	3-Aug	84	88	84	88	84	84	91	84	91	88	97
## 35	4-Aug	91	89	86	89	88	86	91	82	91	91	98
## 36	5-Aug	93	88	90	91	91	88	93	84	90	88	96
## 37	6-Aug	88	84	89	93	91	84	97	82	84	86	89

## 38	7-Aug	91	84	89	93	91	86	87	84	81	80	97
## 39	8-Aug	84	80	86	93	91	88	87	84	82	82	96
## 40	9-Aug	90	73	82	91	96	87	86	86	84	85	95
## 41	10-Aug	89	80	87	90	95	88	88	87	75	83	96
## 42	11-Aug	88	86	88	96	89	86	89	84	82	87	88
## 43	12-Aug	86	88	84	98	89	86	91	81	80	88	84
## 44	13-Aug	84	88	86	97	89	81	91	87	77	86	81
## 45	14-Aug	86	87	80	98	89	87	89	89	82	90	87
## 46	15-Aug	89	88	82	93	94	84	88	90	82	92	86
## 47	16-Aug	90	91	86	93	97	90	90	86	84	89	89
## 48	17-Aug	91	91	84	96	99	91	91	89	86	90	86
## 49	18-Aug	91	89	87	98	101	91	93	90	86	90	88
## 50	19-Aug	90	89	90	98	101	87	91	90	89	89	88
## 51	20-Aug	89	88	79	89	97	86	93	87	88	92	93
## 52	21-Aug	90	82	84	91	87	88	93	88	82	94	91
## 53	22-Aug	91	79	87	91	86	90	91	88	84	93	88
## 54	23-Aug	91	81	87	90	88	88	95	90	84	87	87
## 55	24-Aug	91	82	88	80	92	93	93	89	87	85	83
## 56	25-Aug	84	84	90	82	92	90	91	88	82	84	85
## 57	26-Aug	88	87	91	89	90	91	88	89	86	84	88
## 58	27-Aug	84	90	89	88	90	91	84	90	88	86	88
## 59	28-Aug	86	90	90	90	92	81	82	91	90	86	90
## 60	29-Aug	88	91	93	91	92	86	82	89	87	85	90
## 61	30-Aug	84	91	93	91	88	81	78	88	88	85	88
## 62	31-Aug	82	88	91	84	87	82	77	89	87	85	80
## 63	1-Sep	80	88	87	88	79	80	84	88	82	85	85
## 64	2-Sep	73	91	84	91	81	75	84	86	80	88	86
## 65	3-Sep	87	93	77	84	82	73	89	87	81	87	85
## 66	4-Sep	84	81	90	93	87	81	95	87	82	85	88
## 67	5-Sep	87	81	91	96	81	90	93	84	84	81	83
## 68	6-Sep	89	82	89	96	66	88	91	73	81	81	85
## 69	7-Sep	89	86	90	91	66	87	88	75	86	83	80
## 70	8-Sep	89	88	89	91	75	86	87	81	73	85	83
## 71	9-Sep	91	84	79	77	80	86	91	82	84	86	83
## 72	10-Sep	84	80	78	87	82	89	95	79	84	84	85
## 73	11-Sep	86	82	81	87	84	87	95	80	84	84	84
## 74	12-Sep	88	86	84	87	86	84	90	81	81	86	82
## 75	13-Sep	78	87	89	86	87	84	75	84	79	88	70
## 76	14-Sep	79	87	87	87	86	86	78	82	79	88	80
## 77	15-Sep	86	88	87	89	80	77	91	82	73	91	82
## 78	16-Sep	82	88	88	81	75	77	88	81	75	88	83
## 79	17-Sep	82	90	87	81	73	81	86	81	80	86	85
## 80	18-Sep	78	88	82	82	73	81	81	81	79	88	85
## 81	19-Sep	79	91	80	79	84	82	80	84	78	90	79
## 82	20-Sep	79	95	82	68	87	84	86	87	73	90	73
## 83	21-Sep	78	89	82	79	77	86	84	82	75	90	75
## 84	22-Sep	81	70	88	72	73	87	77	75	80	86	82
## 85	23-Sep	84	80	84	75	81	88	82	81	84	87	86
## 86	24-Sep	84	82	81	78	84	69	73	80	82	88	84
## 87	25-Sep	87	66	82	81	82	66	69	82	81	85	75
## 88	26-Sep	84	70	84	82	68	72	75	82	79	77	78
## 89	27-Sep	79	64	87	78	71	75	75	82	72	86	79
## 90	28-Sep	75	68	80	80	75	78	79	73	78	85	81
## 91	29-Sep	72	77	75	77	73	71	73	66	78	85	70

## 92	30-Sep	64	86	75	71	75	71	79	71	80	82	75
## 93	1-Oct	66	75	86	73	77	75	82	72	82	83	83
## 94	2-Oct	72	73	78	75	79	80	84	68	82	85	81
## 95	3-Oct	84	75	77	84	82	81	84	66	80	83	82
## 96	4-Oct	70	78	82	71	81	80	82	77	81	85	84
## 97	5-Oct	66	81	82	73	82	79	87	78	80	81	86
## 98	6-Oct	64	82	73	71	73	70	86	75	75	72	76
## 99	7-Oct	60	82	82	73	66	68	80	73	75	72	72
## 100	8-Oct	78	82	69	73	55	79	71	73	73	73	72
## 101	9-Oct	70	80	72	72	55	66	66	73	71	70	79
## 102	10-Oct	72	82	73	72	64	73	70	73	71	77	80
## 103	11-Oct	69	82	78	73	71	75	78	66	77	82	80
## 104	12-Oct	69	79	78	70	73	78	84	78	73	74	71
## 105	13-Oct	73	80	78	64	75	78	79	78	64	77	62
## 106	14-Oct	79	68	75	75	75	75	68	78	63	78	69
## 107	15-Oct	81	63	79	73	77	75	57	69	62	79	70
## 108	16-Oct	80	57	78	77	80	62	66	72	71	76	59
## 109	17-Oct	82	66	77	80	80	60	64	68	75	75	71
## 110	18-Oct	66	64	78	71	80	64	68	70	73	81	77
## 111	19-Oct	63	69	82	66	73	71	71	75	68	83	76
## 112	20-Oct	68	70	75	60	73	75	73	78	71	83	69
## 113	21-Oct	79	70	73	64	75	79	71	84	73	80	69
## 114	22-Oct	81	62	63	73	79	80	64	78	73	67	70
## 115	23-Oct	69	63	63	57	75	81	59	78	70	70	53
## 116	24-Oct	73	62	72	59	75	79	68	73	73	56	56
## 117	25-Oct	73	75	75	64	78	73	60	73	78	54	55
## 118	26-Oct	75	71	79	69	75	64	68	68	79	61	62
## 119	27-Oct	75	57	79	75	78	51	69	64	81	63	66
## 120	28-Oct	81	55	79	73	80	55	75	57	78	62	63
## 121	29-Oct	82	64	78	72	75	63	75	70	75	64	72
## 122	30-Oct	82	66	82	75	77	72	68	77	78	69	73
## 123	31-Oct	81	60	79	75	78	71	60	75	82	70	68

##	X2007	X2008	X2009	X2010	X2011	X2012	X2013	X2014	X2015
## 1	95	85	95	87	92	105	82	90	85
## 2	85	87	90	84	94	93	85	93	87
## 3	82	91	89	83	95	99	76	87	79
## 4	86	90	91	85	92	98	77	84	85
## 5	88	88	80	88	90	100	83	86	84
## 6	87	82	87	89	90	98	83	87	84
## 7	82	88	86	94	94	93	79	89	90
## 8	82	90	82	97	94	95	88	90	90
## 9	89	89	84	96	91	97	88	90	91
## 10	86	87	84	90	92	95	87	87	93
## 11	85	89	86	93	95	90	80	85	92
## 12	87	93	90	90	95	84	87	90	93
## 13	86	85	84	91	97	90	78	89	92
## 14	84	88	89	91	90	90	85	90	90
## 15	81	89	89	94	80	90	86	86	89
## 16	86	89	90	89	85	92	87	83	88
## 17	89	88	88	87	87	93	91	86	93
## 18	89	90	82	83	89	93	87	82	92
## 19	88	91	80	90	94	91	90	85	91
## 20	86	94	82	91	91	84	86	76	93
## 21	86	95	86	94	92	90	87	82	93

## 22	79	92	84	95	94	95	85	83	92
## 23	82	87	87	97	92	97	84	88	88
## 24	87	88	88	94	92	97	86	87	91
## 25	87	89	90	95	90	98	89	88	90
## 26	87	87	92	95	94	98	86	89	91
## 27	90	90	90	93	94	97	82	92	92
## 28	89	93	89	90	90	97	86	90	94
## 29	87	92	85	94	93	94	86	82	93
## 30	92	90	82	95	96	96	90	84	94
## 31	90	88	85	95	96	88	80	85	93
## 32	92	89	89	96	91	94	87	81	89
## 33	92	92	83	84	96	99	89	84	94
## 34	94	91	90	92	97	94	88	88	94
## 35	97	91	92	95	85	87	90	90	97
## 36	96	92	92	93	96	90	88	89	95
## 37	98	94	89	93	93	86	88	92	88
## 38	98	90	91	91	93	84	86	95	88
## 39	100	86	92	93	94	92	83	90	92
## 40	103	85	93	94	91	88	89	89	93
## 41	103	85	93	94	95	87	90	86	94
## 42	100	88	95	95	94	85	90	83	91
## 43	90	81	86	95	95	88	90	88	90
## 44	100	81	90	96	95	91	89	84	89
## 45	99	84	90	89	94	88	83	85	90
## 46	102	87	90	90	88	85	73	87	90
## 47	101	86	88	90	90	91	67	88	90
## 48	101	85	87	91	92	87	66	89	89
## 49	97	86	88	93	94	87	77	89	88
## 50	95	90	90	92	96	84	82	86	89
## 51	96	90	88	93	93	84	84	89	88
## 52	99	85	88	93	94	88	84	92	89
## 53	104	82	85	94	98	84	88	93	92
## 54	98	78	81	93	92	88	90	93	87
## 55	95	83	86	90	93	86	84	88	89
## 56	94	78	87	89	95	85	82	84	84
## 57	92	83	90	90	99	90	82	86	86
## 58	88	80	83	89	95	90	86	88	85
## 59	88	86	75	87	95	80	90	91	83
## 60	89	89	86	84	93	86	92	92	81
## 61	89	89	79	85	90	80	87	88	74
## 62	86	88	79	89	92	89	90	89	84
## 63	84	81	71	90	95	91	90	90	87
## 64	83	85	78	91	96	89	84	90	90
## 65	88	83	79	92	95	85	90	92	89
## 66	91	85	83	84	80	77	89	82	92
## 67	89	88	83	85	78	85	89	89	87
## 68	85	87	85	90	75	85	88	91	85
## 69	86	89	84	91	69	92	88	90	85
## 70	88	90	87	93	73	88	91	84	84
## 71	89	88	84	92	81	83	90	84	87
## 72	89	87	80	94	84	84	89	86	85
## 73	89	83	75	96	86	83	89	90	86
## 74	86	87	81	89	87	81	90	92	78
## 75	85	86	80	86	89	81	87	86	75

```
## 76      81      88      82      91      92      83      82      78      77
## 77      82      79      79      91      86      87      84      80      80
## 78      76      80      82      89      72      86      89      86      79
## 79      78      69      73      95      79      83      79      86      83
## 80      79      82      80      93      77      79      78      85      83
## 81      82      81      74      92      77      81      84      84      87
## 82      81      79      81      96      82      79      86      83      89
## 83      78      75      79      95      86      85      73      87      77
## 84      86      84      84      92      80      87      82      82      76
## 85      83      82      83      91      83      81      82      77      81
## 86      89      78      85      88      82      78      71      78      74
## 87      87      82      87      93      88      82      67      77      67
## 88      84      80      85      76      86      86      78      74      71
## 89      85      77      80      81      84      88      79      78      71
## 90      85      86      83      76      79      86      77      74      75
## 91      81      86      72      79      84      84      76      71      77
## 92      79      86      74      76      78      72      77      84      85
## 93      80      74      76      79      65      75      82      86      71
## 94      82      74      75      78      68      72      82      85      66
## 95      77      80      76      68      75      74      82      78      66
## 96      80      83      74      67      80      82      85      65      70
## 97      81      83      62      70      83      82      84      71      73
## 98      82      82      71      73      81      83      84      78      76
## 99      83      82      79      81      79      68      74      82      81
## 100     83      72      80      82      78      63      72      86      82
## 101     81      75      85      85      72      70      76      86      81
## 102     81      77      74      86      68      73      80      86      71
## 103     67      78      77      86      65      75      79      86      73
## 104     72      77      66      80      73      79      81      85      76
## 105     74      77      73      80      74      75      82      85      81
## 106     78      80      66      73      77      77      77      75      78
## 107     78      81      61      78      80      77      68      69      81
## 108     76      83      61      76      84      74      74      70      77
## 109     82      69      51      80      85      75      72      80      70
## 110     77      67      55      78      80      74      73      76      66
## 111     76      65      61      82      67      73      63      73      64
## 112     75      66      68      77      59      71      70      73      71
## 113     78      72      71      80      63      76      72      77      76
## 114     72      68      74      78      68      79      69      70      79
## 115     81      62      72      76      70      78      63      72      81
## 116     59      54      69      81      73      79      66      74      76
## 117     61      67      65      76      76      80      56      77      71
## 118     68      70      65      85      77      80      61      84      67
## 119     67      59      60      76      79      70      69      84      56
## 120     70      50      71      74      74      56      64      77      78
## 121     62      59      75      68      59      56      75      73      70
## 122     67      65      66      71      61      56      78      68      70
## 123     71      67      69      75      65      65      74      63      62
```

```
data = as.vector(unlist(temps[,2:21]))
temps_ts = ts(data, start = 1996, frequency = 123)
temps_ts
```

```
## Time Series:
```

```

## Start = c(1996, 1)
## End = c(2015, 123)
## Frequency = 123
##      [1] 98 97 97 90 89 93 93 91 93 93 90 91 93 93 82 91 96 95
##     [19] 96 99 91 95 91 93 84 84 82 79 90 91 87 86 90 84 91 93
##     [37] 88 91 84 90 89 88 86 84 86 89 90 91 91 90 89 90 91 91
##     [55] 91 84 88 84 86 88 84 82 80 73 87 84 87 89 89 89 91 84
##     [73] 86 88 78 79 86 82 82 78 79 79 78 81 84 84 87 84 79 75
##     [91] 72 64 66 72 84 70 66 64 60 78 70 72 69 69 73 79 81 80
##    [109] 82 66 63 68 79 81 69 73 73 75 75 81 82 82 81 86 90 93
##    [127] 91 84 84 75 87 84 87 84 88 86 90 91 91 89 89 89 90 89
##    [145] 84 87 88 89 89 91 91 89 88 72 80 84 88 89 88 84 84 80
##    [163] 73 80 86 88 88 87 88 91 91 89 89 88 82 79 81 82 84 87
##    [181] 90 90 91 91 88 88 91 93 81 81 82 86 88 84 80 82 86 87
##    [199] 87 88 88 90 88 91 95 89 70 80 82 66 70 64 68 77 86 75
##    [217] 73 75 78 81 82 82 82 80 82 82 79 80 68 63 57 66 64 69
##    [235] 70 70 62 63 62 75 71 57 55 64 66 60 91 88 91 91 91 89
##    [253] 93 95 95 91 91 86 88 87 91 87 90 91 95 91 91 89 91 91
##    [271] 86 88 80 88 89 90 86 86 82 84 86 90 89 89 86 82 87 88
##    [289] 84 86 80 82 86 84 87 90 79 84 87 87 88 90 91 89 90 93
##    [307] 93 91 87 84 77 90 91 89 90 89 79 78 81 84 89 87 87 88
##    [325] 87 82 80 82 82 88 84 81 82 84 87 80 75 75 86 78 77 82
##    [343] 82 73 82 69 72 73 78 78 78 75 79 78 77 78 82 75 73 63
##    [361] 63 72 75 79 79 79 78 82 79 84 82 87 88 90 91 82 86 87
##    [379] 87 82 77 73 81 81 86 82 87 88 90 90 91 93 93 91 93 93
##    [397] 93 93 97 99 96 93 88 89 91 93 93 93 91 90 96 98 97 98
##    [415] 93 93 96 98 98 89 91 91 90 80 82 89 88 90 91 91 84 88
##    [433] 91 84 93 96 96 91 91 77 87 87 87 86 87 89 81 81 82 79
##    [451] 68 79 72 75 78 81 82 78 80 77 71 73 75 84 71 73 71 73
##    [469] 73 72 72 73 70 64 75 73 77 80 71 66 60 64 73 57 59 64
##    [487] 69 75 73 72 75 75 89 91 93 95 96 96 96 91 96 99 96 93
##    [505] 91 93 93 93 91 97 100 99 93 96 87 82 75 82 88 91 89 87
##    [523] 86 86 81 84 88 91 91 91 91 96 95 89 89 89 89 94 97 99
##    [541] 101 101 97 87 86 88 92 92 90 90 92 92 88 87 79 81 82 87
##    [559] 81 66 66 75 80 82 84 86 87 86 80 75 73 73 84 87 77 73
##    [577] 81 84 82 68 71 75 73 75 77 79 82 81 82 73 66 55 55 64
##    [595] 71 73 75 75 77 80 80 80 73 73 75 79 75 75 78 75 78 80
##    [613] 75 77 78 84 87 87 84 86 87 87 89 91 87 90 90 86 82 82
##    [631] 84 87 88 90 87 84 87 90 84 82 88 90 84 89 89 87 84 84
##    [649] 84 86 88 84 86 88 87 88 86 86 81 87 84 90 91 91 87 86
##    [667] 88 90 88 93 90 91 91 81 86 81 82 80 75 73 81 90 88 87
##    [685] 86 86 89 87 84 84 86 77 77 81 81 82 84 86 87 88 69 66
##    [703] 72 75 78 71 71 75 80 81 80 79 70 68 79 66 73 75 78 78
##    [721] 75 75 62 60 64 71 75 79 80 81 79 73 64 51 55 63 72 71
##    [739] 90 90 87 89 93 93 89 89 90 91 84 77 82 88 91 93 93 93
##    [757] 93 91 95 91 89 87 84 86 89 91 91 88 90 93 91 91 91 93
##    [775] 97 87 87 86 88 89 91 91 89 88 90 91 93 91 93 93 91 95
##    [793] 93 91 88 84 82 82 78 77 84 84 89 95 93 91 88 87 91 95
##    [811] 95 90 75 78 91 88 86 81 80 86 84 77 82 73 69 75 75 79
##    [829] 73 79 82 84 84 82 87 86 80 71 66 70 78 84 79 68 57 66
##    [847] 64 68 71 73 71 64 59 68 60 68 69 75 75 68 60 73 81 87
##    [865] 86 80 84 87 90 89 84 84 86 87 84 86 88 88 88 88 88 89
##    [883] 86 81 82 84 87 87 89 88 84 88 84 84 84 82 84 82 84 84
##    [901] 86 87 84 81 87 89 90 86 89 90 90 87 88 88 90 89 88 89

```

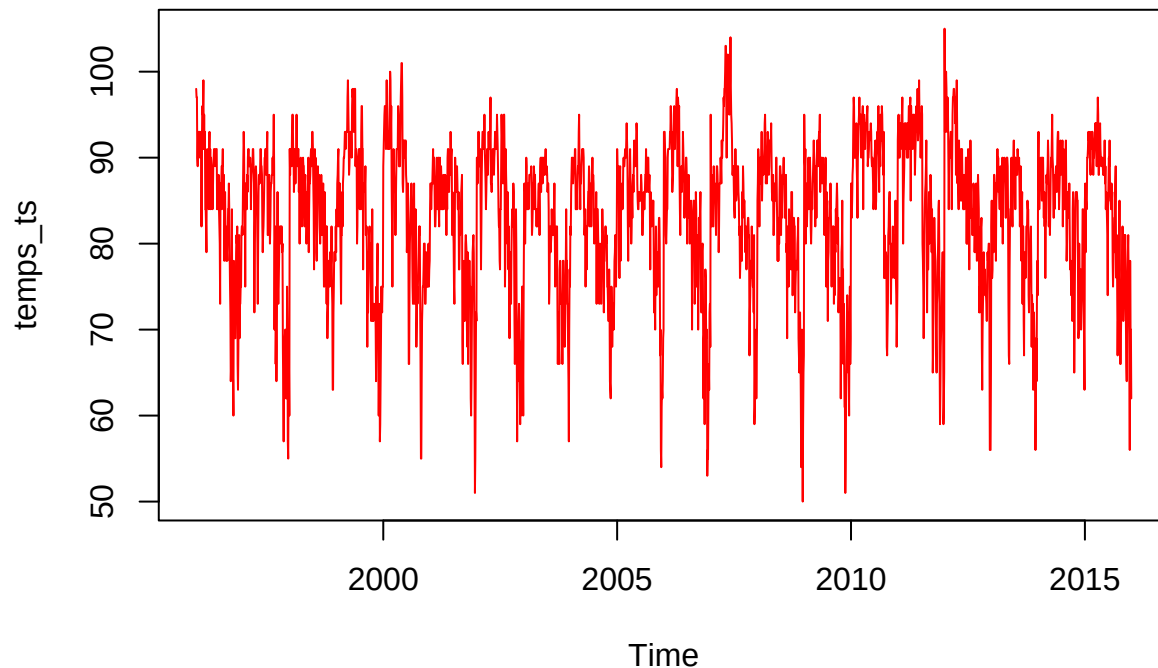
##	[919]	90	91	89	88	89	88	86	87	87	84	73	75	81	82	79	80	81	84
##	[937]	82	82	81	81	81	84	87	82	75	81	80	82	82	82	73	66	71	72
##	[955]	68	66	77	78	75	73	73	73	73	66	78	78	78	69	72	68	70	75
##	[973]	78	84	78	78	73	73	68	64	57	70	77	75	82	81	86	88	90	90
##	[991]	89	87	88	89	90	89	91	91	84	84	84	87	84	88	89	89	93	95
##	[1009]	89	87	84	89	87	89	90	91	90	91	91	90	84	81	82	84	75	82
##	[1027]	80	77	82	82	84	86	86	89	88	82	84	84	87	82	86	88	90	87
##	[1045]	88	87	82	80	81	82	84	81	86	73	84	84	84	81	79	79	73	75
##	[1063]	80	79	78	73	75	80	84	82	81	79	72	78	78	80	82	82	80	81
##	[1081]	80	75	75	73	71	71	77	73	64	63	62	71	75	73	68	71	73	73
##	[1099]	70	73	78	79	81	78	75	78	82	91	89	86	86	89	82	76	88	89
##	[1117]	78	83	86	84	87	84	85	89	90	89	89	90	91	91	90	92	94	92
##	[1135]	90	83	78	84	82	86	88	91	88	86	80	82	85	83	87	88	86	90
##	[1153]	92	89	90	90	89	92	94	93	87	85	84	84	86	86	85	85	85	85
##	[1171]	88	87	85	81	81	83	85	86	84	84	86	88	88	91	88	86	88	90
##	[1189]	90	90	86	87	88	85	77	86	85	85	82	83	85	83	85	81	72	72
##	[1207]	73	70	77	82	74	77	78	79	76	75	81	83	83	80	67	70	56	54
##	[1225]	61	63	62	64	69	70	93	93	93	91	90	81	80	82	84	84	90	91
##	[1243]	91	91	91	91	93	93	96	93	93	91	86	87	88	93	95	96	91	91
##	[1261]	94	95	95	97	98	96	89	97	96	95	96	88	84	81	87	86	89	86
##	[1279]	88	88	93	91	88	87	83	85	88	88	90	90	88	80	85	86	85	88
##	[1297]	83	85	80	83	83	85	84	82	70	80	82	83	85	85	79	73	75	82
##	[1315]	86	84	75	78	79	81	70	75	83	81	82	84	86	76	72	72	79	80
##	[1333]	80	71	62	69	70	59	71	77	76	69	69	70	53	56	55	62	66	63
##	[1351]	72	73	68	95	85	82	86	88	87	82	82	89	86	85	87	86	84	81
##	[1369]	86	89	89	88	86	86	79	82	87	87	87	90	89	87	92	90	92	92
##	[1387]	94	97	96	98	98	100	103	103	100	90	100	99	102	101	101	97	95	96
##	[1405]	99	104	98	95	94	92	88	88	89	89	86	84	83	88	91	89	85	86
##	[1423]	88	89	89	89	86	85	81	82	76	78	79	82	81	78	86	83	89	87
##	[1441]	84	85	85	81	79	80	82	77	80	81	82	83	83	81	81	67	72	74
##	[1459]	78	78	76	82	77	76	75	78	72	81	59	61	68	67	70	62	67	71
##	[1477]	85	87	91	90	88	82	88	90	89	87	89	93	85	88	89	89	88	90
##	[1495]	91	94	95	92	87	88	89	87	90	93	92	90	88	89	92	91	91	92
##	[1513]	94	90	86	85	85	88	81	81	84	87	86	85	86	90	90	85	82	78
##	[1531]	83	78	83	80	86	89	89	88	81	85	83	85	88	87	89	90	88	87
##	[1549]	83	87	86	88	79	80	69	82	81	79	75	84	82	78	82	80	77	86
##	[1567]	86	86	74	74	80	83	83	82	82	72	75	77	78	77	77	80	81	83
##	[1585]	69	67	65	66	72	68	62	54	67	70	59	50	59	65	67	95	90	89
##	[1603]	91	80	87	86	82	84	84	86	90	84	89	89	90	88	82	80	82	86
##	[1621]	84	87	88	90	92	90	89	85	82	85	89	83	90	92	92	89	91	92
##	[1639]	93	93	95	86	90	90	90	88	87	88	90	88	88	85	81	86	87	90
##	[1657]	83	75	86	79	79	71	78	79	83	83	85	84	87	84	80	75	81	80
##	[1675]	82	79	82	73	80	74	81	79	84	83	85	87	85	80	83	72	74	76
##	[1693]	75	76	74	62	71	79	80	85	74	77	66	73	66	61	61	51	55	61
##	[1711]	68	71	74	72	69	65	65	60	71	75	66	69	87	84	83	85	88	89
##	[1729]	94	97	96	90	93	90	91	91	94	89	87	83	90	91	94	95	97	94
##	[1747]	95	95	93	90	94	95	95	96	84	92	95	93	93	91	93	94	94	95
##	[1765]	95	96	89	90	90	91	93	92	93	93	94	93	90	89	90	89	87	84
##	[1783]	85	89	90	91	92	84	85	90	91	93	92	94	96	89	86	91	91	89
##	[1801]	95	93	92	96	95	92	91	88	93	76	81	76	79	76	79	78	68	67
##	[1819]	70	73	81	82	85	86	86	80	80	73	78	76	80	78	82	77	80	78
##	[1837]	76	81	76	85	76	74	68	71	75	92	94	95	92	90	90	94	94	91
##	[1855]	92	95	95	97	90	80	85	87	89	94	91	92	94	92	92	90	94	94
##	[1873]	90	93	96	96	91	96	97	85	96	93	93	94	91	95	94	95	95	94

```

## [1891] 88 90 92 94 96 93 94 98 92 93 95 99 95 95 93 90 92 95
## [1909] 96 95 80 78 75 69 73 81 84 86 87 89 92 86 72 79 77 77
## [1927] 82 86 80 83 82 88 86 84 79 84 78 65 68 75 80 83 81 79
## [1945] 78 72 68 65 73 74 77 80 84 85 80 67 59 63 68 70 73 76
## [1963] 77 79 74 59 61 65 105 93 99 98 100 98 93 95 97 95 90 84
## [1981] 90 90 90 92 93 93 91 84 90 95 97 97 98 98 97 97 94 96
## [1999] 88 94 99 94 87 90 86 84 92 88 87 85 88 91 88 85 91 87
## [2017] 87 84 84 88 84 88 86 85 90 90 80 86 80 89 91 89 85 77
## [2035] 85 85 92 88 83 84 83 81 81 83 87 86 83 79 81 79 85 87
## [2053] 81 78 82 86 88 86 84 72 75 72 74 82 82 83 68 63 70 73
## [2071] 75 79 75 77 77 74 75 74 73 71 76 79 78 79 80 80 70 56
## [2089] 56 56 65 82 85 76 77 83 83 79 88 88 87 80 87 78 85 86
## [2107] 87 91 87 90 86 87 85 84 86 89 86 82 86 86 90 80 87 89
## [2125] 88 90 88 88 86 83 89 90 90 90 89 83 73 67 66 77 82 84
## [2143] 84 88 90 84 82 82 86 90 92 87 90 90 84 90 89 89 88 88
## [2161] 91 90 89 89 90 87 82 84 89 79 78 84 86 73 82 82 71 67
## [2179] 78 79 77 76 77 82 82 82 85 84 84 74 72 76 80 79 81 82
## [2197] 77 68 74 72 73 63 70 72 69 63 66 56 61 69 64 75 78 74
## [2215] 90 93 87 84 86 87 89 90 90 87 85 90 89 90 86 83 86 82
## [2233] 85 76 82 83 88 87 88 89 92 90 82 84 85 81 84 88 90 89
## [2251] 92 95 90 89 86 83 88 84 85 87 88 89 89 86 89 92 93 93
## [2269] 88 84 86 88 91 92 88 89 90 90 92 82 89 91 90 84 84 86
## [2287] 90 92 86 78 80 86 86 85 84 83 87 82 77 78 77 74 78 74
## [2305] 71 84 86 85 78 65 71 78 82 86 86 86 86 85 85 75 69 70
## [2323] 80 76 73 73 77 70 72 74 77 84 84 77 73 68 63 85 87 79
## [2341] 85 84 84 90 90 91 93 92 93 92 90 89 88 93 92 91 93 93
## [2359] 92 88 91 90 91 92 94 93 94 93 89 94 94 97 95 88 88 92
## [2377] 93 94 91 90 89 90 90 90 89 88 89 88 89 92 87 89 84 86
## [2395] 85 83 81 74 84 87 90 89 92 87 85 85 84 87 85 86 78 75
## [2413] 77 80 79 83 83 87 89 77 76 81 74 67 71 71 75 77 85 71
## [2431] 66 66 70 73 76 81 82 81 71 73 76 81 78 81 77 70 66 64
## [2449] 71 76 79 81 76 71 67 56 78 70 70 62

```

```
plot(temps_ts, col = "red")
```



```
temps_holt = HoltWinters(temps_ts, seasonal = "multiplicative")
summary(temps_holt)
```

```
##           Length Class  Mode
## fitted      9348   mts    numeric
## x           2460   ts     numeric
## alpha         1  -none-  numeric
## beta          1  -none-  numeric
## gamma         1  -none-  numeric
## coefficients  125  -none-  numeric
## seasonal      1  -none-  character
## SSE           1  -none-  numeric
## call          3  -none-  call
```

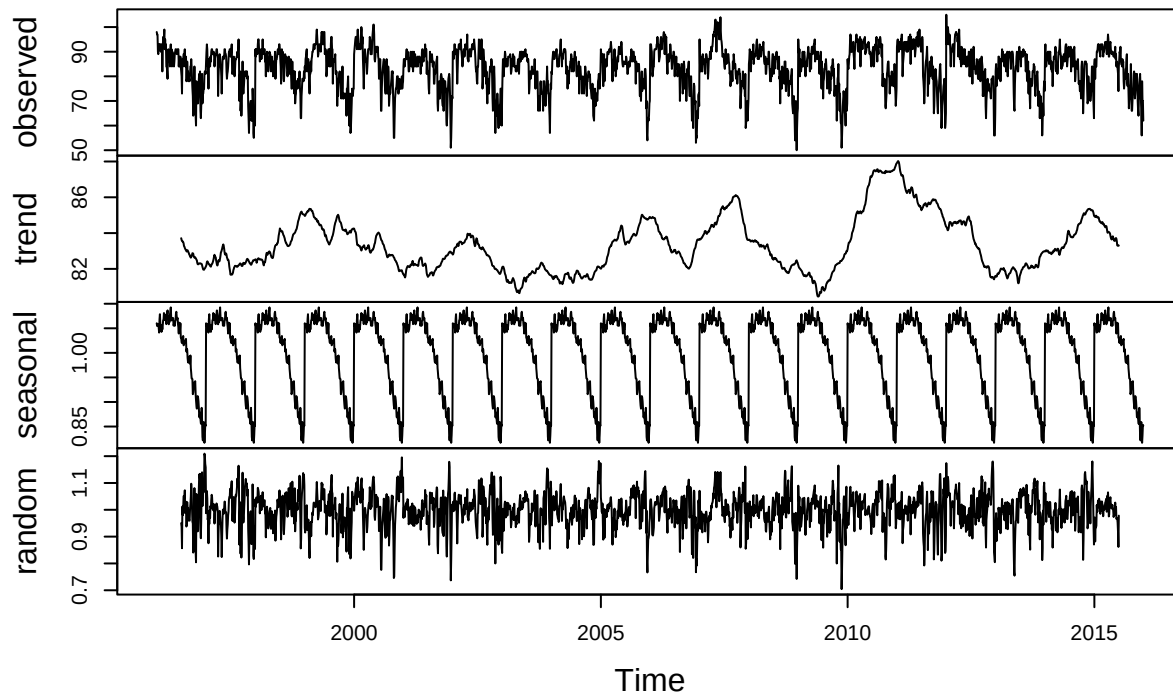
```
temps_holt$fitted[1:20,]
```

```
##           xhat    level    trend    season
## [1,] 87.23653 82.87739 -0.004362918 1.0526529
## [2,] 90.42182 82.15059 -0.004362918 1.1007422
## [3,] 92.99734 81.91055 -0.004362918 1.1354128
## [4,] 90.94030 81.90763 -0.004362918 1.1103378
## [5,] 83.99917 81.93634 -0.004362918 1.0252306
## [6,] 84.04496 81.93247 -0.004362918 1.0258379
## [7,] 75.06333 81.90115 -0.004362918 0.9165601
## [8,] 87.04945 81.85429 -0.004362918 1.0635250
## [9,] 84.02220 81.82134 -0.004362918 1.0269532
## [10,] 87.06445 81.80368 -0.004362918 1.0643665
## [11,] 84.05272 81.76208 -0.004362918 1.0280709
## [12,] 88.05141 81.72617 -0.004362918 1.0774531
## [13,] 86.03178 81.69246 -0.004362918 1.0531740
## [14,] 89.92787 81.66954 -0.004362918 1.1011777
```

```
## [15,] 90.89514 81.70546 -0.004362918 1.1125326
## [16,] 90.93680 81.75907 -0.004362918 1.1123128
## [17,] 88.91874 81.78965 -0.004362918 1.0872217
## [18,] 88.89374 81.83125 -0.004362918 1.0863634
## [19,] 88.86666 81.88704 -0.004362918 1.0852925
## [20,] 89.83703 81.95824 -0.004362918 1.0961900
```

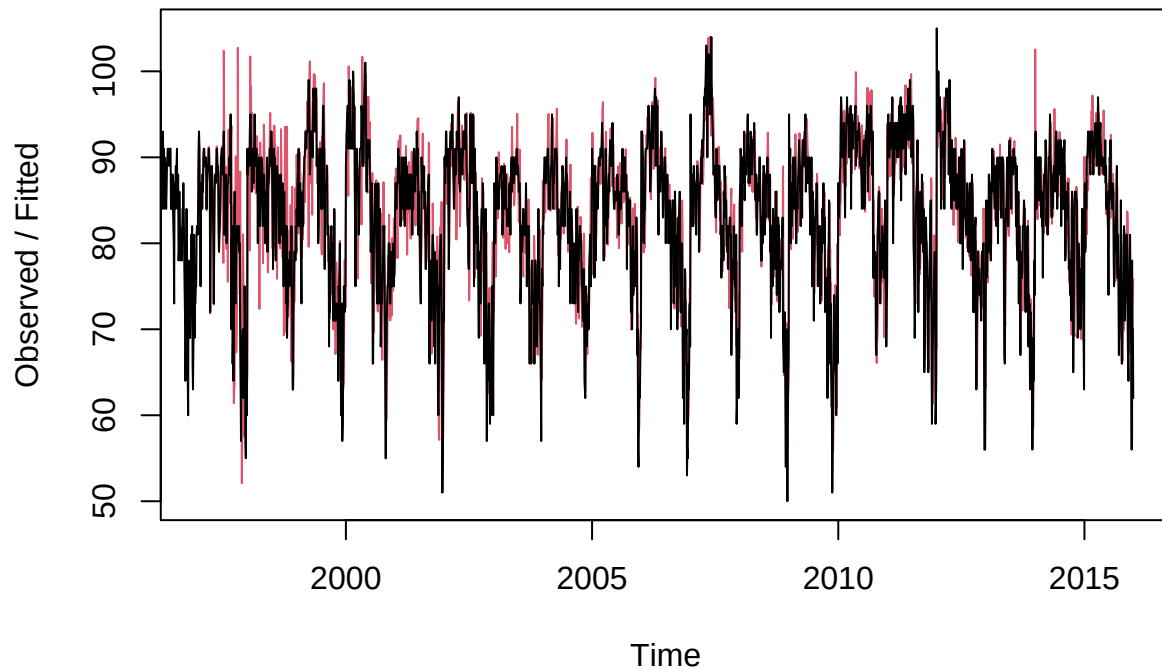
```
plot(decompose(temps_ts, type = "multiplicative"))
```

Decomposition of multiplicative time series

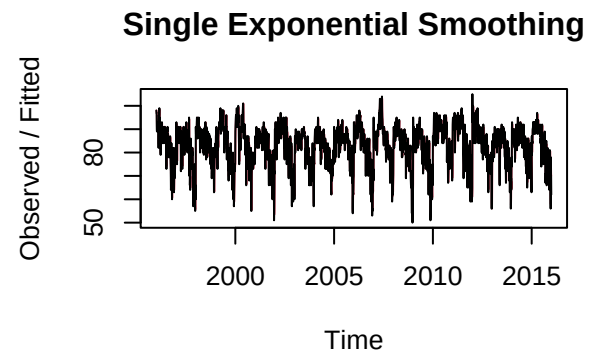
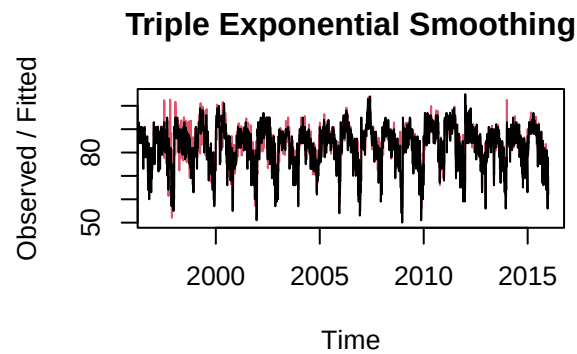


```
plot(temps_holt)
```

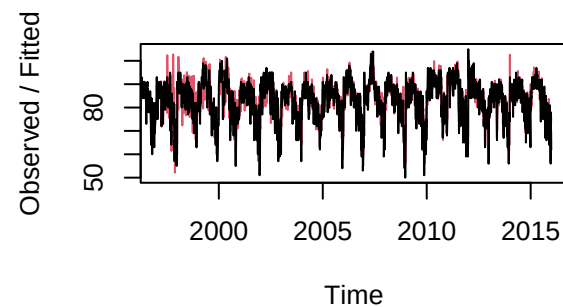
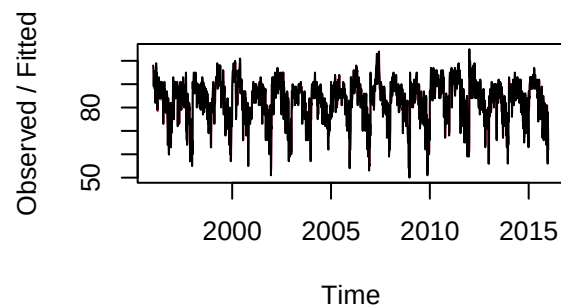
Holt-Winters filtering



```
temps_holt1 <- HoltWinters(temps_ts, alpha = NULL, beta = FALSE, gamma = FALSE, seasonal = "multiplicat  
temps_holt2 <- HoltWinters(temps_ts, alpha = NULL, beta = NULL, gamma = FALSE, seasonal = "multiplicati  
  
## Warning in HoltWinters(temps_ts, alpha = NULL, beta = NULL, gamma = FALSE, :  
## optimization difficulties: ERROR: ABNORMAL_TERMINATION_IN_LNSRCH  
  
temps_holt3 <- HoltWinters(temps_ts, alpha = NULL, beta = FALSE, gamma = NULL, seasonal = "multiplicati  
  
par(mfrow=c(2,2))  
plot(temps_holt, main="Triple Exponential Smoothing")  
plot(temps_holt1, main="Single Exponential Smoothing")  
plot(temps_holt2, main="Double Exponential Smoothing without Seasonality")  
plot(temps_holt3, main="Double Exponential Smoothing without Trend")
```



Double Exponential Smoothing without Seasonality Double Exponential Smoothing without Trend



```
temp_smoothed_mx <- matrix(temps_holt$fitted[,1], nrow = 123)

temp_smoothed <- as.data.frame(temp_smoothed_mx)

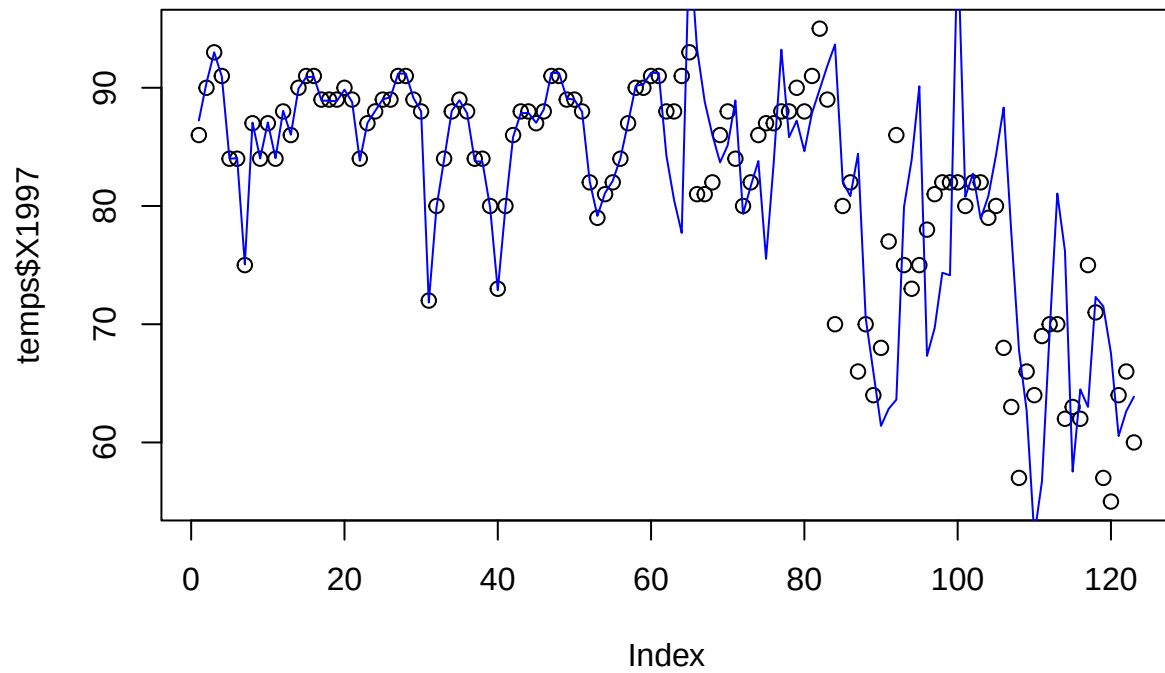
colnames(temp_smoothed) <- colnames(temps)[c(3:21)]

temp_smoothed$DAY <- temps$DAY
head(temp_smoothed)
```

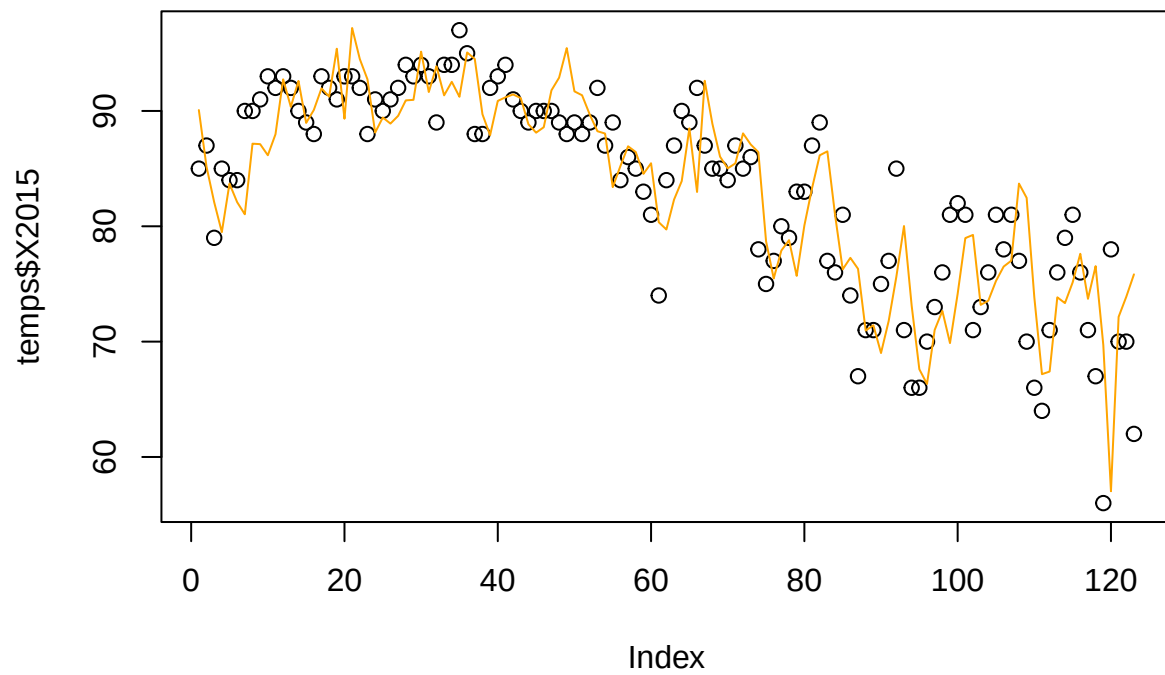
```
##      X1997      X1998      X1999      X2000      X2001      X2002      X2003      X2004
## 1 87.23653 65.04516 90.29613 83.39938 87.68863 78.07509 73.10059 87.27074
## 2 90.42182 84.87634 85.44878 86.44444 84.78855 86.02384 72.13247 85.01878
## 3 92.99734 89.61560 85.65942 92.85774 88.70570 90.23022 77.77739 82.68648
## 4 90.94030 88.47600 84.80741 91.55309 86.98750 87.27931 83.52416 83.37312
## 5 83.99917 83.11178 81.14293 88.80208 81.40681 86.06745 83.86090 83.64904
## 6 84.04496 88.00054 85.21673 91.04477 81.83758 87.87757 78.93483 86.79140
##      X2005      X2006      X2007      X2008      X2009      X2010      X2011      X2012
## 1 92.29714 78.50826 81.58696 84.72917 79.51855 86.74604 93.88371 82.30605
## 2 92.85614 88.18138 88.52648 80.39548 85.65722 81.47324 87.43846 92.55001
## 3 92.33884 92.43570 86.72311 84.53380 88.31357 82.29310 90.24836 91.18746
## 4 87.29596 92.69774 83.30574 89.62822 88.56597 82.90566 93.69353 95.13130
## 5 84.25223 90.58916 84.18954 89.27001 89.12501 80.92784 90.14667 94.60910
## 6 85.75665 86.91496 82.21750 84.28967 79.32562 84.52016 88.67069 96.75445
##      X2013      X2014      X2015      DAY
## 1 84.88750 102.54643 90.07756 1-Jul
## 2 76.18707 89.57468 85.16854 2-Jul
## 3 81.46207 88.15080 82.09161 3-Jul
## 4 76.56780 87.11605 79.49314 4-Jul
```

```
## 5 75.42085 85.20682 83.69666 5-Jul  
## 6 78.42462 83.25423 82.08838 6-Jul
```

```
plot(temps$X1997)  
lines(temp_smoothed$X1997, col = "blue")
```



```
plot(temps$X2015)  
lines(temp_smoothed$X2015, col = "orange")
```



```
library(forecast)
```

```
## Warning: package 'forecast' was built under R version 4.3.3
```

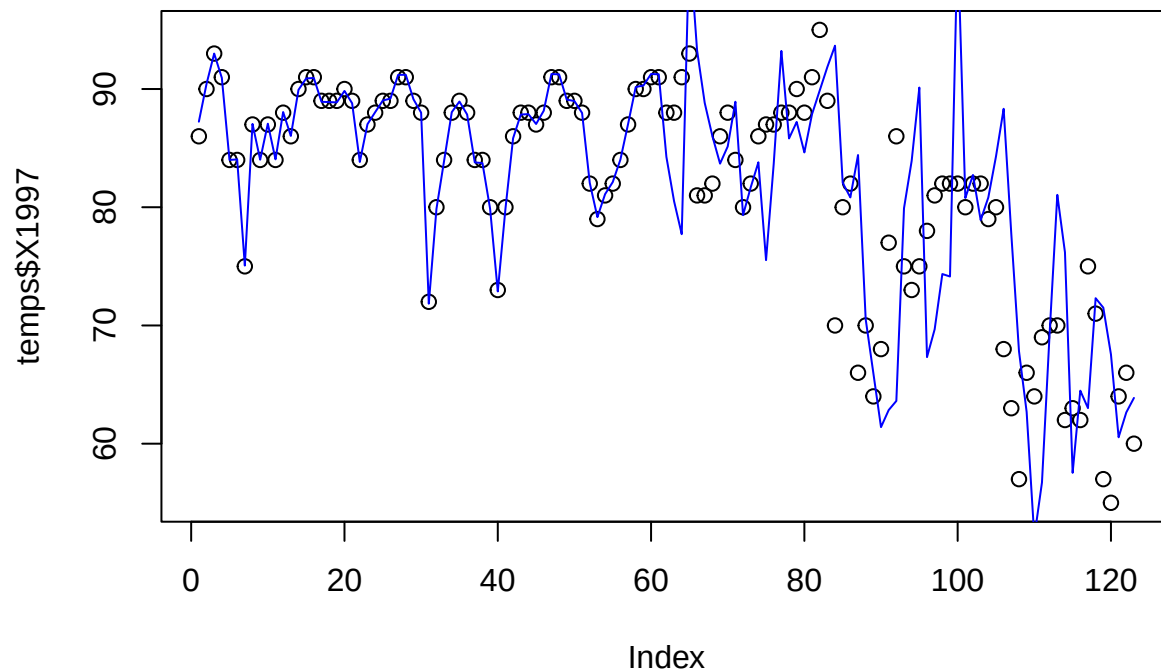
```
## Registered S3 method overwritten by 'quantmod':
```

```
##   method          from
```

```
## as.zoo.data.frame zoo
```

```
plot(temps$X1997)
```

```
lines(temp_smoothed$X1997, col = "blue")
```



```
s <- rep(0,123)
```

```
#initialize the first value as 0
```

```
s[[1]] <- 0
```

```
#start with a C value of 1/2 of the standard deviation of the year 1997
```

```
sd(temp_smoothed$X1997)
```

```
## [1] 9.698415
```

```
c0 <- 0.5*sd(temp_smoothed$X1997)
```

```
#calculate the mean value of the observed proper "summer" high temperature, or the first 75 days of dai  
mean_temp0 <- mean(temp_smoothed$X1997[1:75])
```

```
#calculate the s values by day.
```

```
#Note that we are detecting the change in a decreasing time series trend.
```

```
for (i in 2:123){
```

```
  s[[i]] <- max(0, s[[i-1]] + (mean_temp0 - temp_smoothed$X1997[i] - c0))
```

```
}
```

```
s
```

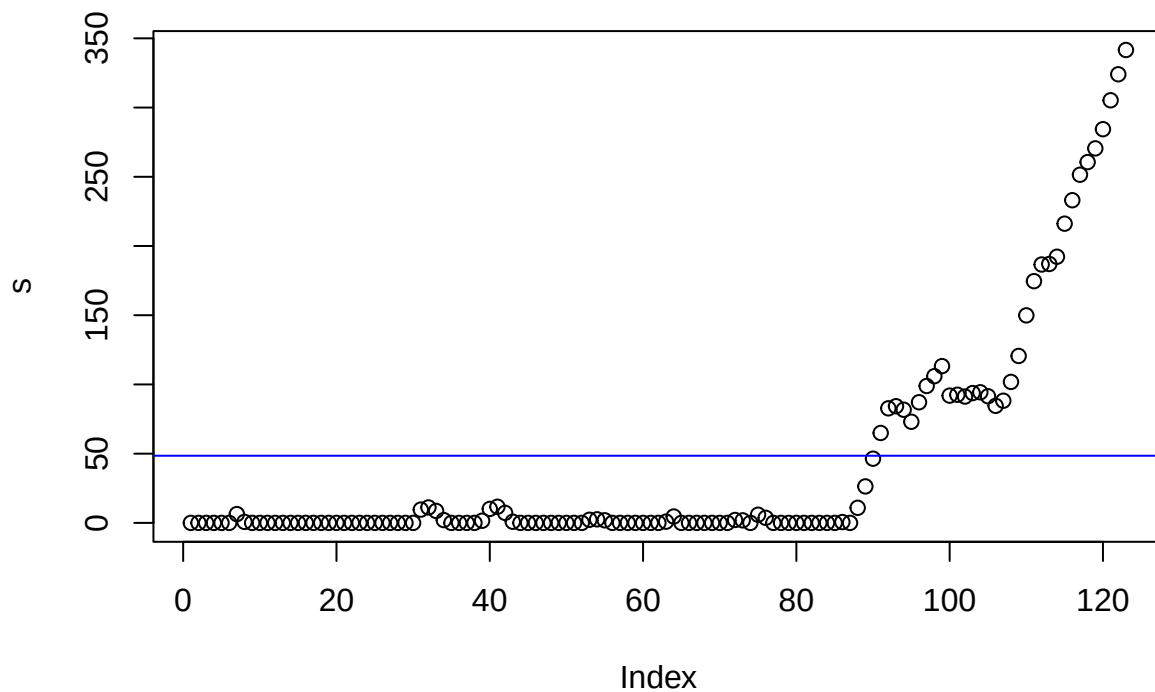
```
## [1] 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
## [7] 6.3569122 0.7277125 0.0000000 0.0000000 0.0000000 0.0000000
## [13] 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
## [19] 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
## [25] 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
## [31] 9.5794382 11.1429493 8.6338517 2.0091768 0.0000000 0.0000000
## [37] 0.0000000 0.0000000 1.5534967 10.0958373 11.6500171 7.2393701
## [43] 0.8166848 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
## [49] 0.0000000 0.0000000 0.0000000 0.0000000 2.2552507 2.5667872
## [55] 1.8600863 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
## [61] 0.0000000 0.0000000 0.9029190 4.6028355 0.0000000 0.0000000
## [67] 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 2.0896444
## [73] 1.8255843 0.0000000 5.8952088 3.6541254 0.0000000 0.0000000
## [79] 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
## [85] 0.0000000 0.5886107 0.0000000 10.8894314 26.3466950 46.3663100
## [91] 64.9433113 82.7484974 84.2545548 81.7470046 73.0435032 87.1417993
## [97] 98.8850876 105.9539415 113.2338339 91.9211392 92.5600852 91.2504245
## [103] 93.7515819 94.3456749 91.4629503 84.5541915 88.2401684 101.8331409
## [109] 120.5662720 149.8946680 174.5923693 186.6351219 187.0064757 192.2371797
## [115] 216.1244110 233.0593262 251.4704299 260.5828192 270.4964667 284.3694599
## [121] 305.2487992 324.0237981 341.5733703
```

```
t0 <- 5*sd(temp_smoothed[,1])
```

```
#plot out the s values over time and the t threshold value
```

```
plot(s)
```

```
abline(h=t0, col="blue")
```



```
min(which(s>=t0))
```

```
## [1] 91
```

```
#store all the s values by year in a matrix. It should have 20 columns, including the first column record
s_all <- data.frame(matrix(ncol = 20, nrow = 123))
colnames(s_all) <- colnames(temps)[c(1,3:21)]
```

```
#The function
```

```
summer_end <- function(i,c,t){

  s_all[,i][1] <- 0

  temp_mean <- mean(temp_smoothed[,i][1:75])

  for (j in 2:123){
    s_all[,i][j] <- max(0, s_all[,i][j-1] + temp_mean - temp_smoothed[,i][j] - c)
  }

  return (min(which(s_all[,i]>=t)))
}
```

```
#Store the detected end of summer day for each year from 1997 to 2015.
```

```
end_summer <- rep(0,19)

for (i in 1:19){
  end_summer[[i]] <- summer_end(i,0.5*sd(temp_smoothed[,i]),5*sd(temp_smoothed[,i]))
}

end_summer
```

```
## [1] 91 93 86 71 91 88 92 80 100 84 82 87 93 93 72 86 90 88 81
```

```
#store the actual dates in a new vector
```

```
end_summer_dates <- vector(length(end_summer), mode = "list")
#assign the date
for (i in 1:19){
  end_summer_dates[[i]] <- temp_smoothed$DAY[end_summer[[i]]]
}

head(end_summer_dates)
```

```
## [[1]]
## [1] "29-Sep"
##
## [[2]]
## [1] "1-Oct"
##
## [[3]]
## [1] "24-Sep"
##
## [[4]]
## [1] "9-Sep"
##
## [[5]]
```

```
## [1] "29-Sep"  
##  
## [[6]]  
## [1] "26-Sep"
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3
```

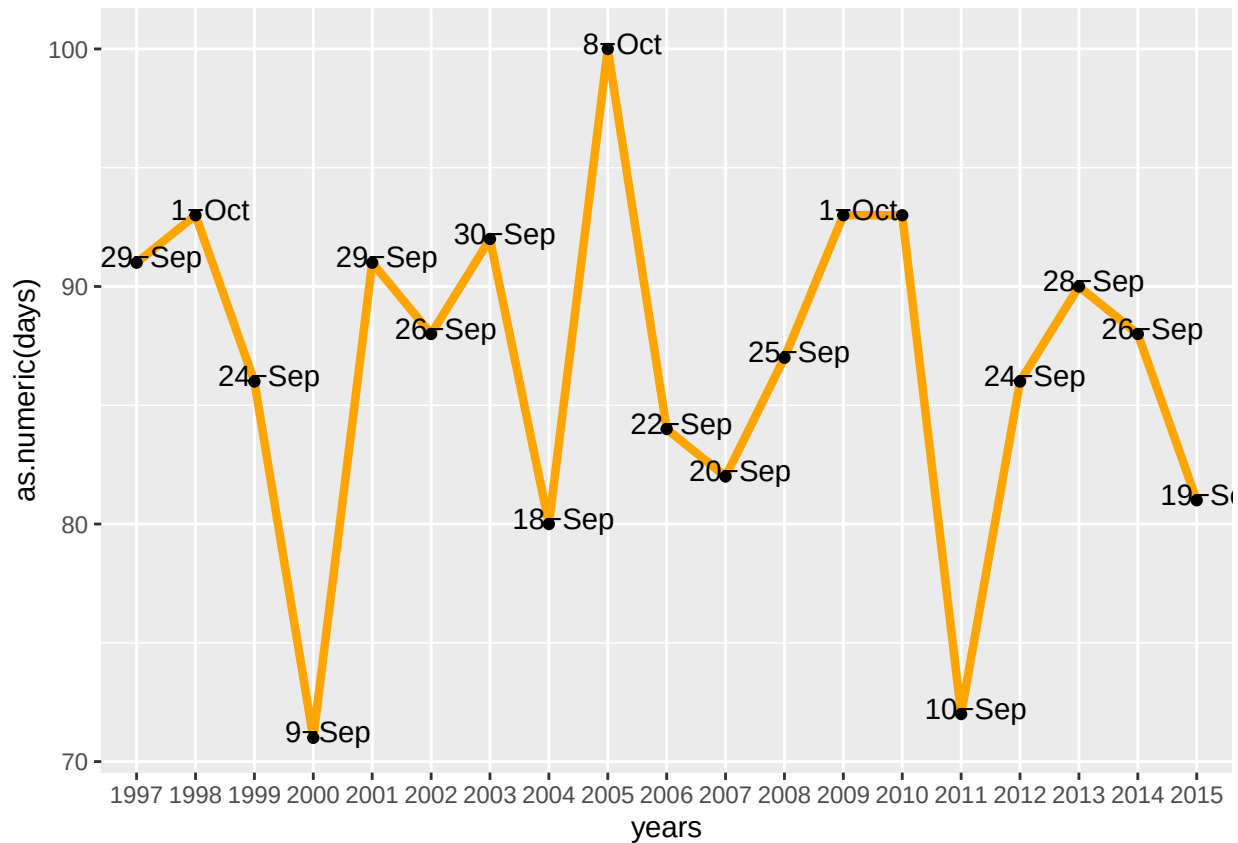
```
#get a clean year column  
years <- sub('.', '', colnames(temp_smoothed)[1:19])  
years
```

```
## [1] "1997" "1998" "1999" "2000" "2001" "2002" "2003" "2004" "2005" "2006"  
## [11] "2007" "2008" "2009" "2010" "2011" "2012" "2013" "2014" "2015"
```

```
#create a data frame for plotting  
end_summer_df <- data.frame(cbind(year = years, days = as.numeric(end_summer)))  
end_summer_df
```

```
##   year days  
## 1  1997   91  
## 2  1998   93  
## 3  1999   86  
## 4  2000   71  
## 5  2001   91  
## 6  2002   88  
## 7  2003   92  
## 8  2004   80  
## 9  2005  100  
## 10 2006   84  
## 11 2007   82  
## 12 2008   87  
## 13 2009   93  
## 14 2010   93  
## 15 2011   72  
## 16 2012   86  
## 17 2013   90  
## 18 2014   88  
## 19 2015   81
```

```
#plot the detected end of summer day by year  
ggplot(data=end_summer_df, aes(x=years, y=as.numeric(days), group=1)) +  
  geom_line(color="orange", lwd=1.5) +  
  geom_point() +  
  geom_text(  
    label=end_summer_dates,  
    nudge_x = 0.25, nudge_y = 0.25,  
    check_overlap = T  
  )
```



```
#Same process but using the unsmoothed data
s_all0 <- data.frame(matrix(ncol = 21, nrow = 123))
colnames(s_all0) <- colnames(temps)[1:21]

summer_end0 <- function(i,c,t){

  s_all0[,i][1] <- 0

  temp_mean <- mean(temps[,i][1:75])

  for (j in 2:123){
    s_all0[,i][j] <- max(0, s_all0[,i][j-1] + temp_mean - temps[,i][j] - c)
  }

  return (min(which(s_all0[,i]>=t)))
}

end_summer0 <- rep(0,20)

for (i in 2:21){
  end_summer0[[i-1]] <- summer_end0(i,0.5*sd(temps[,i]),5*sd(temps[,i]))
}

end_summer_dates0 <- vector(length(end_summer0), mode = "list")
#assign the date
for (i in 1:20){
```

```

    end_summer_dates0[[i]] <- temps$DAY[end_summer0[[i]]]
  }

years <- sub('.', '', colnames(temps)[3:21])
years

## [1] "1997" "1998" "1999" "2000" "2001" "2002" "2003" "2004" "2005" "2006"
## [11] "2007" "2008" "2009" "2010" "2011" "2012" "2013" "2014" "2015"

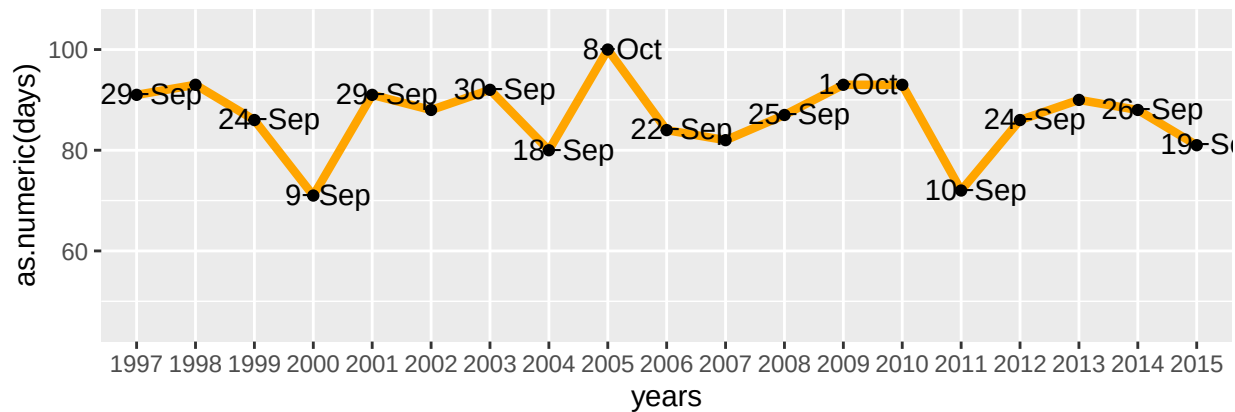
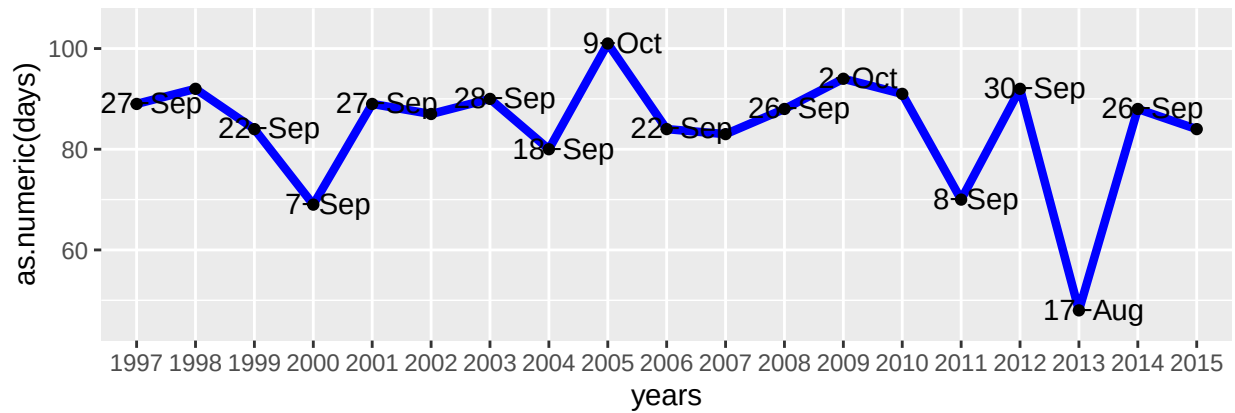
#create a data frame for plotting
end_summer_df0 <- data.frame(cbind(year = years,
                                   days = as.numeric(end_summer0[2:20])))

library(gridExtra)
#Raw data
p1= ggplot(data=end_summer_df0, aes(x=years, y=as.numeric(days), group=1)) +
  geom_line(color="blue",lwd=1.5)+
  geom_point()+
  ylim(45, 105) +
  geom_text(
    label=end_summer_dates0[2:20],
    nudge_x = 0.25, nudge_y = 0.25,
    check_overlap = T
  )

p2 = ggplot(data=end_summer_df, aes(x=years, y=as.numeric(days), group=1)) +
  geom_line(color="orange",lwd=1.5)+
  geom_point()+
  ylim(45, 105)+
  geom_text(
    label=end_summer_dates,
    nudge_x = 0.25, nudge_y = 0.25,
    check_overlap = T
  )

grid.arrange(p1, p2, nrow = 2)

```



```
end_summer
```

```
## [1] 91 93 86 71 91 88 92 80 100 84 82 87 93 93 72 86 90 88 81
```

```
test_df <- data.frame(year = c(1:19), days = as.numeric(end_summer))
#A quick graphical analysis
scatter.smooth(x=test_df$year, y=test_df$days, main="Summer Ending Day ~ Year")
```

Summer Ending Day ~ Year

